

Hidden Emotion Detection Using Voice Signals

Saadhvi S
School of Computer Science and
Engineering
RV University
Bangalore, India
saadhvis.btech23@rvu.edu.in

SathyaShree KV
School of Computer Science and
Engineering
RV University
Bangalore, India
sathyashree.btech23@rvu.edu.in

Shobana S
School of Computer Science and
Engineering
RV University
Bangalore, India
shobanas.btech23@rvu.edu.in

Shree Vidhya B
School of Computer Science and Engineering
RV University
Bangalore, India
shreev.btech23@rvu.edu.in

Abstract—Human emotions play a critical role in communication; nonetheless, these emotions are usually concealed from each other owing to specific personal or societal environmental variables. Even though it may be difficult to perceive human emotions through written texts or facial expressions, specific emotions exist in speech, including pitch, tone, intensity, and rhythm. Nevertheless, it is somewhat difficult to use speech to detect emotions since emotions vary according to speaker properties, accent, recording circumstances, and noise. Several speech-based emotion recognition models have failed to efficiently capture both temporal and spectral characteristics of speech data. Emotions are dynamic in nature, meaning that they are time-dependent and can shift depending on the circumstances. The distinction between highly correlated emotions like fear and anger increases complexity. Several impacts are associated with inefficiency in detecting emotions, including psychological disease treatment, human-computer communication, and customer care services. Efficient emotion recognition techniques will allow for improved user experience, early psychophysical condition diagnosis, and well-informed choices. The research proposes a speech emotion recognition framework that leverages the HuBERT model and machine learning techniques to enhance the performance of emotion classification. The proposed system will learn meaningful representations of speech and improve the recognition of dynamic emotional patterns in speech.

I. INTRODUCTION

Human communication includes more than just words; it also involves emotions. These emotions are often shown in how we speak, including our tone, pitch, rhythm, and the strength of our voice. Concealed emotions in speeches have now become one of the main concerns in SER. In this project, a model based on HuBERT, along with self-supervised and supervised learning techniques, is applied to discover hidden emotions in speech signal data.

The latest trends in emotion recognition consist of transformer-based architectures, self-supervised learning techniques, and multimodality. Some of the best examples of models that provide strong speech representations in large-scale unlabeled data are HuBERT, wav2vec2, and emotion2vec. Attention mechanisms and feature fusions are effective in capturing both temporal and spectral parts of the signal.

Transformer models help us capture global and local features. Multimodal approaches consist of speech, text, and facial expression analysis, thus enhancing emotion recognition. Speech emotion recognition faces challenges because of speaker variations, noise interferences, and multiple co-occurring emotions.

In this project, the motivation is understanding the hidden emotions that are not expressed openly through communication. Voice signal analysis using machine learning algorithms will help capture and analyze emotions. Emotion detection systems can be used in many domains and applications.

II. RELATED WORK

Zhang et al. trained models on labeled and unlabeled datasets such as IVD, AVD, and IEMOCAP, leveraging MMD and MixMatch techniques. The model was able to reach an F1-score of 0.414 on IVD and also registered a higher weighted accuracy on IEMOCAP, which illustrates the strength of curriculum learning for large unlabeled data [1].

BIG-FUSION is an approach proposed by Poria et al. that incorporates global and local contextual features with dual-attention transformer architecture along with graph contrastive learning. It scores a weighted averaging F1-score of 67.17% for both the IEMOCAP and MELD databases, demonstrating superior performance compared to others because of the diverse features used in training and the reduction in contextual bias [2].

Emotion2vec is an approach developed by Ma et al., which involves a self-supervised pretraining process using extremely large databases with over 932k samples. The hybrid CNN-transformer achieves up to 77.64% accuracy, showing a great level of generality [3].

DialogueRNN is an attention-based recurrent neural network designed for dialogue emotion classification. Using the IEMOCAP database, it obtains an F1 score of 62.75%, outperforming other best works by 6.62% [4].

Liu et al. implemented an AI-generated music-based emotion regulation system through EEG signals. The system integrates CNN and diffusion models and reports 90.6% accuracy for binary classification. On the downside, the system is constrained by the limited amount of data available [5].

Abdelwahab and Busso presented ArabEmoNet, a hybrid CNN-BiLSTM model that utilized mel-spectrograms and attention mechanisms. Their model demonstrated a remarkable accuracy of 99.46% on KEDAS and 91.48% on KSUEmotions datasets. [6].

Lotfian and Busso investigated the use of multimodal emotion recognition with FFNN, CNN, and ensemble techniques. Their experimentation revealed that CNN-based architectures are superior to the traditional FFNNs, and their final model hit an accuracy of 74.64% [7].

Deng et al. have come up with a music recommendation system that is aware of emotions. It uses a facial emotion recognition technique along with some deep learning models such as DRViT. This system has quite high correlation scores in measuring valence and arousal [8].

Gulati et al. came up with the Emotion Neural Transducer (ENT), which is a combination of wav2vec2 and an RNN transducer architecture. The weight accuracy of this model hit around 75%, and it was a good one-two punch of speech recognition and emotion detection [9].

Schmitt et al. used a deep learning beamforming approach for speech emotion recognition in noisy environments with MSP-Podcast data; the model achieved 92% accuracy and showed that it was still quite effective while being robust to real-world conditions [10].

Li et al. presented a speech emotion captioning system named SECap that integrates HuBERT, Q-Former, and LLaMA. It can interpret complex emotional states and outperformed target metrics such as SIM1 and MOS scores [11].

Wu et al. put forward a selective acoustic feature enhancement that relies on GAN-based architectures. Their system has significantly improved correlation of arousal and dominance, and thus it is very well suited for noisy speech environments [12].

III. METHODOLOGY

A. System Overview

The proposed model processes raw audio input and produces two outputs: explicit emotion prediction and hidden emotion score. Speech signals are first preprocessed and passed through the HuBERT-based encoder. The encoder output is then refined through self-supervised adapter layers before being fed into a dual-path attention module that generates separate explicit and hidden emotion predictions. The KL divergence between these two predictions is the suppression score.

B. Data Collection and Preprocessing

We used two publicly accessible datasets for emotional speech recognition, taken from the Kaggle platform.

IEMOCAP - Interactive Emotional Dyadic Motion Capture

The IEMOCAP dataset has become one of the most popular

datasets in terms of automatic analysis of speech emotions and consists of about 12 hours of dyadic conversation collected from ten actors in five sessions. Their speech expressions have been annotated with human labeling.

CREMA-D - Crowd-Sourced Emotional Multimodal Actors Dataset

CREMA-D is an extensive audiovisual dataset consisting of 7,442 speech segments collected from 91 actors of different age groups (20-74), ethnic backgrounds, and genders, classified into six emotional categories using 12 fixed sentences.

Data Preprocessing Module

This module preprocesses raw audio recordings from the IEMOCAP and CREMA-D datasets for model consumption. Only four emotion classes, anger, happiness, neutral, and sadness, are retained from the dataset. The frustrated class is removed due to poor inter-annotator agreement and limited representation across sessions. Next, the audios are resampled to 16 KHz and segmented into six-second clips using Wav2Vec Feature Extractor. A weighted random sampler is used to handle class imbalance. Session 5 of IEMOCAP is held out as the test set throughout all stages. CREMA-D audio is separately preprocessed for use in the self-supervised learning stage, where masked hidden state prediction is performed without requiring emotion labels.

C. Implementation

For implementation, we are using Google Colab for GPU access and to speed up the training and inference of the model.

Training Module

i) Stage 1: Supervised Fine-Tuning

First, a pre-trained HuBERT model is fine-tuned on the IEMOCAP dataset for four-class emotion classification. HuBERT is a transformer-based self-supervised model pre-trained on large quantities of unlabelled speech. A classification head is attached to the encoder output, and the entire model is optimised using cross-entropy loss with a label smoothing coefficient of 0.05 to improve generalisation. The trained weights are saved as the Stage 1 checkpoint (stage1_best.pt), which is used as the initialisation point for the next stages.

ii) Stage 2: Self-Supervised Adapter Learning

The encoder weights from Stage 1 are frozen, and only the adapter parameters are trained. A lightweight bottleneck adapter layer is introduced into each transformer block of the HuBERT encoder. This parameter-efficient design reduces computational cost while preserving the rich speech representations acquired during pre-training. The adapters are trained on the CREMA-D dataset using a masked hidden state reconstruction task, where a subset of encoder frames is masked, and the model is trained to reconstruct the original hidden states from the surrounding context. This self-supervised objective further enriches the encoder's emotion representations without requiring additional labelled data. The adapter weights are saved separately as stage2_adapters.pt.

iii) Stage 3: Dual-Path Attention Training

The third stage features a dual-path attention module built on a frozen encoder and adapters. This module consists of two parallel attention mechanisms that work on the same sequence of encoder hidden states. The local attention path uses a windowed query-key-value mechanism with a frame window of four, capturing detailed, frame-level micro-expression features. The global attention path uses a single learnable query vector applied across the full utterance, capturing the overall expressed emotion at the utterance level. The output of the local path is passed to a hidden emotion classification head, while the output of the global path is passed to an explicit emotion classification head. Both heads consist of a linear layer, GELU activation, dropout, and a four-class softmax output. The two paths are trained with a composite loss function designed to actively separate their representation

$$Loss = CE_{exp} + CE_{hid} + \lambda_{orth} \cdot cosine_sim - \lambda_{div} \cdot KL$$

CE_{exp} and CE_{hid} are loss functions used for supervising the explicit head and hidden head using the corresponding emotion labels, respectively. The orthogonality loss acts as a penalty for the cosine similarity of global and local representation vectors, which ensures that both paths do not share identical feature representation. The negative KL divergence term with a weight of $\lambda_{div} = 0.3$ is the critical factor in finding hidden emotions: taking the KL divergence of the distributions of the two heads and subtracting the result from the total loss maximizes their difference. This compels

the hidden head to explore an alternate way to interpret the emotion behind the same utterance, thereby achieving emotional suppression. The best checkpoint is saved as `stage3_best_fixed.pt`.

Inference Module

The inference module loads the Stage 3 checkpoint and processes unseen utterances to produce two outputs per utterance: an explicit emotion prediction from the global attention head, and a hidden emotion suppression score derived from the KL divergence between the two heads' softmax probability distributions. A suppression threshold is computed as the mean KL score plus one standard deviation across the test set. Utterances whose KL score exceeds this threshold are flagged as samples likely containing suppressed emotion.

Validation Module

The validation module evaluates the correctness of the suppression detector without requiring explicit hidden emotion labels, which do not exist in any publicly available dataset. Instead, it uses inter-annotator disagreement entropy from the IEMOCAP dataset as a ground-truth proxy for suppressed emotion. IEMOCAP utterances are annotated independently by multiple evaluators, and those utterances where there is significant disagreement among the annotators are considered emotionally ambiguous — in line with the theory that concealed or suppressed emotions lead to unclear perceived emotions. The module computes the Shannon entropy of the annotator vote distribution for each utterance. Utterances with entropy above the median are labelled as high-disagreement (ambiguous), and those below as low-disagreement (clear). The KL suppression scores produced by the inference module are then evaluated against this binary label using three complementary checks:

- Check A — Detection Validity: AUC-ROC and Mann-Whitney U test to assess whether the KL score is statistically higher on ambiguous utterances.
- Check B — Head Disagreement: Comparison of the rate at which the explicit and hidden heads predict different emotion classes on high- versus low-entropy utterances.
- Check C — Distribution Shift: Analysis of whether the hidden head's emotion distribution differs systematically on ambiguous utterances versus clear ones.

IV. RESULTS AND DISCUSSION

Experimental results were obtained with respect to the IEMOCAP Session 5 dataset and external speech data. The evaluation consisted of two independent test sessions: Session A – IEMOCAP Session 5 Dataset with Annotator Disagreement Analysis

Session B – Testing on External Speech Samples

Evaluation was performed for both tasks of explicit emotion classification and hidden emotion detection implemented in the proposed architecture.

As previously mentioned, the HuBERT model was tested on Session 5 of the IEMOCAP dataset. It was noted that there could be multiple opinions among human annotators in this dataset, which can be considered as evidence of emotional suppression. Therefore, the proposed model was able to perform not only explicit emotion classification but also hidden emotion detection by means of measuring the discrepancy between explicit emotions and hidden emotions through the KL divergence.

	precision	recall	f1-score	support
angry	0.69	0.71	0.70	170
happy	0.70	0.60	0.65	442
neutral	0.62	0.73	0.67	384
sad	0.73	0.69	0.71	245
accuracy			0.67	1241
macro avg	0.68	0.68	0.68	1241
weighted avg	0.68	0.67	0.67	1241

IEMOCAP Dataset evaluation results

In particular, the HuBERT-based three-stage architecture showed good classification performance with an overall accuracy of 67% over the four classes of emotions, namely angry, happy, neutral, and sad.

For hidden emotion detection, the model uses the KL divergence to measure the level of divergence between the explicit emotion branch and the hidden emotion branch. In general, higher KL divergence values indicate more disagreement between explicit and hidden emotions, which can be a sign of emotional suppression.

```

=== Hidden Emotion Analysis ===
Mean KL divergence      : 0.0257
Std KL divergence      : 0.0869
Detection threshold     : 0.1125
Likely suppressed      : 55/1241 (4.4%)
Head disagreement overall: 0.053
Head disagree (flagged) : 0.455
Head disagree (unflagged): 0.035

=== Suppression Patterns (expressed -> hidden, flagged utterances) ===
neutral -> angry      : 5 (9.1%)
angry -> happy       : 5 (9.1%)
happy -> angry       : 3 (5.5%)
angry -> neutral    : 3 (5.5%)
sad -> neutral      : 2 (3.6%)
sad -> happy       : 2 (3.6%)

```

Hidden Emotion Analysis and Suppression Patterns

Annotator Disagreement Analysis (IEMOCAP Session 5)

To assess the reliability of ambiguous samples and validate the hidden emotion detection signals, an annotator disagreement analysis was conducted on the IEMOCAP Session 5 test set (n=1219).

Detection Validity (Check A)

Entropy-based ambiguity flagging revealed 263 (21.6%) of 1219 utterances as high-entropy (ambiguous). The Mann-Whitney test indicated that there was a statistically significant difference in KL divergence scores between high- and low-entropy groups ($U=144,681$; $p = 0.0001$) based on the mean KL (high=0.0240; low=0.0265). Unfortunately, the AUC-ROC of 0.5754 only provides a poor discriminatory signal and indicates that entropy is only a modest but statistically valid predictor of annotated ambiguity. The best results ($P=0.272$; $R=0.316$; $F1=0.292$) regarding precision and recall for threshold-based flagging strategies occurred at the 75th percentile ($KL>0.0176$) and outperformed both the 90th percentile ($F1=0.156$) and mean+1std ($F1=0.063$) thresholds.

```

=====
CHECK A - DETECTION VALIDITY
=====
Entropy threshold : 0.0000
Positive (ambiguous): 263/1219 (21.6%)
KL>mean+1std (0.1136): Flagged=55 P=0.182 R=0.038 F1=0.063
KL>75th pct (0.0176): Flagged=305 P=0.272 R=0.316 F1=0.292
KL>90th pct (0.0470): Flagged=122 P=0.246 R=0.114 F1=0.156

AUC-ROC: 0.5754 ~ Weak signal
Mann-Whitney: U=144681, p=0.0001 | Mean KL high=0.0240 low=0.0265
✓ Significant at p<0.05

```

Detection Validity Check

Explicit vs. Hidden Head Disagreement (Check B)

The inconsistency rate between the visible and the non-visible attention heads was much greater during ambiguous (high-entropy) than during clear (low-entropy) messages (0.072 vs 0.047). This finding supports the hypothesis that ambiguous speech generates a greater amount of internal divergence in the two heads' representations of the messages. Of the 19 "smoking gun" messages in which the suppression patterns were most evident, the most frequent suppression pattern was from happy to neutral (31.6%), followed by from sad to neutral (21.1%), indicating that both positive (happy) and negative (sad) emotional states tend to be suppressed most often with a neutral representation. There was a small number of instances of suppression patterns of angry to happy (10.5%), angry to neutral (10.5%), neutral to angry (10.5%), and neutral to sad (10.5%).

```

=====
CHECK B - EXPLICIT vs HIDDEN HEAD DISAGREEMENT
=====
Head disagree on HIGH entropy: 0.072
Head disagree on LOW entropy: 0.047
✓ Heads disagree more on ambiguous

Smoking gun utterances: 19
Top suppression patterns (expressed -> hidden):
happy -> neutral : 6 (31.6%)
sad -> neutral : 4 (21.1%)
angry -> happy : 2 (10.5%)
angry -> neutral : 2 (10.5%)
neutral -> angry : 2 (10.5%)
neutral -> sad : 2 (10.5%)

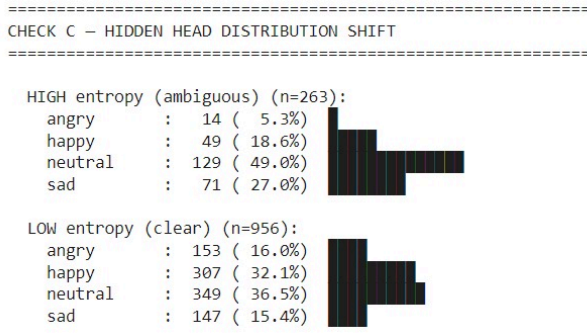
```

Explicit VS Hidden Head Disagreement Check

Hidden Head Distribution Shift (Check C)

The data reveal that there was a large distributional difference in the predictions made for hidden heads of utterances that contained high- and low-entropy emotions. The most frequently predicted emotion (in any instance of ambiguously expressed emotion), out of the total number of emotions, was the neutral emotion (49%), followed by sadness (27.0%), happiness (18.6%), and only 5.3% being identified as angry. In comparison to the high-entropy emotion data, the predictions made for low-entropy emotions exhibited a relatively equal distribution of emotional predictions among the four categories (36.5% neutral; 32.1% happy; 16.0% angry; 15.4% sad). The greater percentage of neutral predictions for the ambiguous samples lends credence to the assertion that emotions that have been suppressed or felt

little/no emotion will be reflected in the hidden representation, as was noted in Check B suppression data.



Hidden Head Distribution Shift Check

Discussion

Researchers used a HuBERT-based, three-tiered framework to identify hidden emotion in speech through: (1) fine-tuning using supervised learning; (2) creating self-supervised adapted learning by aggregating data across multiple sessions; and (3) using a dual-path attention approach (separation of processing streams based on the multiple modalities utilized). An experimental data set from IEMOCAP was then captured and utilized in experiments to evaluate the stability of performance and the ability to detect instances of negative emotion expression suppressed from high affect verbalization through measuring disagreement via Kullback-Leibler (KL) divergence. Cross-dataset experiments confirmed that the framework has a high degree of generalization ability for previously unseen external samples of speech.

Overall, this three-tier framework for identifying emotion in conversational speech will provide researchers with effective tools for evaluating emotional inconsistencies. Additionally, this framework has strong potential for being integrated as a tool for the analysis of hidden emotions.

ACKNOWLEDGMENT

We would like to express our sincere gratitude for your mentorship, Prof. Subir Roy, in the School of Computer Science and Engineering, R V University, Bangalore.

REFERENCES

- [1] Z. Zhang, F. Ringeval, and B. Schuller, “Inferring Emotion from Large-Scale Internet Voice Data: A Semi-Supervised Curriculum Augmentation Based Deep Learning Approach,” in *Proc. Interspeech*, 2020.
- [2] S. Poria, E. Cambria, and A. Hussain, “BIG-FUSION: Brain-Inspired Global-Local Context Fusion Framework for Multimodal Emotion Recognition in Conversations,” in *Proc. ACL*, 2021.
- [3] H. Ma et al., “emotion2vec: Self-Supervised Pre-Training for Speech Emotion Representation,” in *Proc. Interspeech*, 2023.
- [4] N. Majumder, S. Poria, D. Hazarika, and E. Cambria, “DialogueRNN: An Attentive RNN for Emotion Detection in Conversations,” in *Proc. AAAI Conf. Artificial Intelligence*, 2019.
- [5] Y. Liu et al., “Enhancing Emotion Regulation in Mental Disorder Treatment: An AI GC-Based Closed-Loop Music Intervention System,” in *IEEE Access*, 2022.
- [6] A. A. Abdelwahab and M. Busso, “ArabEmoNet: A Lightweight Hybrid CNN-BiLSTM Model with Attention for Arabic Speech Emotion Recognition,” in *Proc. ICASSP*, 2021.
- [7] R. Lotfian and C. Busso, “Classifying Emotional Utterances by Employing Multi-Modal Speech Emotion Recognition,” *IEEE Trans. Affective Computing*, vol. 10, no. 3, pp. 1–10, 2019.
- [8] S. Deng et al., “Emotion-Aware Music Recommendation,” in *Proc. ACM Multimedia*, 2020.
- [9] A. Gulati et al., “Emotion Neural Transducer for Fine-Grained Speech Emotion Recognition,” in *Proc. ICASSP*, 2022.
- [10] M. Schmitt et al., “Speech Emotion Recognition with Deep Learning Beamforming on a Distant Human-Robot Interaction Scenario,” in *Proc. Interspeech*, 2017.
- [11] Y. Li et al., “SECap: Speech Emotion Captioning with Large Language Model,” in *Proc. ACL*, 2023.

[12] J. Wu et al., “Selective Acoustic Feature Enhancement for Speech Emotion Recognition With Noisy Speech,” *IEEE Signal Processing Letters*, vol. 29, pp. 1–5, 2022.



0% Overall Similarity

Filters

Hidden Emotion Detection Using Voice Signals

Sudhish S
School of Computer Science and
Engineering
RV University
Bangalore, India
sudhishs@rvv.edu.in

Sathyadinesh RV
School of Computer Science and
Engineering
RV University
Bangalore, India
sathyadinesh@rvv.edu.in

Shubham S
School of Computer Science and
Engineering
RV University
Bangalore, India
shubham@rvv.edu.in

Shiva Vallaya H
School of Computer Science and Engineering
RV University
Bangalore, India
shivavallaya@rvv.edu.in

Abstract—Human emotions play a critical role in communication. Nonetheless, these emotions are usually concealed from each other owing to specific personal or societal environmental variables. Even though it may be difficult to perceive human emotions through written texts or facial expressions, specific emotions exist in speech, including pitch, tone, intensity, and rhythm. Nevertheless, it is somewhat difficult to use speech to detect emotions since emotions vary according to speaker properties, accent, recording circumstances, and noise. Several speech-based emotion recognition models have failed to efficiently capture both temporal and spectral characteristics of speech data. Emotions are dynamic in nature, meaning that they are time-dependent and can shift depending on the circumstances. The distinction between highly correlated emotions like fear and anger increases complexity. Several aspects are associated with inefficiency in detecting emotions, including psychological distress treatment, human-computer communication, and customer care services. Efficient emotion recognition techniques will allow for improved user experience, early psychological condition diagnosis, and well-informed choices. The research proposes a speech emotion recognition framework that leverages the HuBERT model and machine learning techniques to enhance the performance of emotion classification. The proposed system will learn meaningful representations of speech and improve the recognition of dynamic emotional patterns in speech.

I. INTRODUCTION

Human communication includes more than just words; it also involves emotions. These emotions are often shown in how we speak, including our tone, pitch, rhythm, and the strength of our voice. Concealed emotions in speeches have now become one of the main concerns in SER. In this project, a model based on HuBERT along with self-supervised and supervised learning techniques is applied to discover hidden emotions in speech signal data. The latest trends in emotion recognition involve: (i) transformer-based architectures, self-supervised learning techniques, and multimodality. Some of the best examples of models that provide strong speech representations in large-scale unlabeled data are HuBERT, wv2vec2, and emotion2vec. Attention mechanisms and feature fusion are effective in capturing both temporal and spectral parts of the signal. Transformer models help in capturing global and local features. Multimodal approaches consist of speech, text, and facial expression analysis, thus enhancing emotion recognition. Speech emotion recognition faces challenges because of speaker variations, noise interference, and multiple co-occurring emotions. In this project, the motivation is understanding the hidden emotions that are not expressed openly through communication. Voice signal analysis using machine learning algorithms will help capture and analyze emotions. Emotion detection systems can be used in many domains and applications.

II. RELATED WORK

Zhang et al. studied models on labeled and unlabeled datasets such as IVD, AVD, and IEMOCAP, leveraging MMD and Midlevel techniques. The model was able to reach an F1 score of 0.414 on IVD and also registered a higher weighted accuracy on IEMOCAP which illustrates the strength of automatic learning for large unlabeled data [1].

BKG-FUSION is an approach proposed by Jena et al. that incorporates global and local contextual features with dual-attention mechanism, achieving along with graph convolutional learning. It scores a weighted average F1 score of 67.17% for both the IEMOCAP and MELD datasets, demonstrating superior performance compared to others because of the diverse features used in training and the reduction in contextual bias [2].

EmotionVec is an approach developed by Ma et al., which involves a self-supervised pretraining process using extremely large datasets with over 80k samples. The hybrid 176k transformer achieves up to 77.60% accuracy, showing a great level of generality [3].

DialogGNN is an attention-based recurrent neural network designed for dialogue emotion classification. Using the IEMOCAP dataset, it achieves an F1 score of 83.75%, surpassing other best works by 5.82% [4].

Lin et al. implemented an AI-generated music-based emotion regulation system through EEG signals. The system integrates GNN and diffusion models and reports 90.6% accuracy for binary classification. On the demands, the system is constrained by the limited amount of data available [5].

Abdelwahab and Basse presented AudioNet, a hybrid CNN-BLSTM model that utilized mel-spectrograms and attention mechanisms. Their model demonstrated a remarkable accuracy of 99.66% on KRDAS and 91.48% on KS3Emotions datasets [6].

Luffian and Basse investigated the use of multimodal emotion recognition with FNN, CNN, and generative techniques. Their experimentation revealed that 176k-based architectures are superior to the traditional FNNs, and their first model hit an accuracy of 74.64% [7].

Deng et al. have come up with a music recommendation system that is aware of emotions. It uses a local emotion recognition technique along with some deep learning models such as GNN. This system has quite high correlation scores in measuring valence and arousal [8].

Gulati et al. came up with the Emotion Neural Transformer (ENTL) which is a combination of wv2vec2 and an RNN transformer architecture. The weighted accuracy of this model hit around 75%, and it was a great machine punch of speech recognition and emotion detection [9].

Schmitt et al. used a deep learning framework approach for speech emotion recognition in noisy environments with MSP-Pollux data; the model achieved 93% accuracy and showed that it was still quite effective while being robust to real-world scenarios [10].

Li et al. presented a speech emotion capturing system named SP-Cap that integrates HuBERT, Q-former, and LLaMA. It can interpret complex emotional states and outputted target metrics such as SDI and MOI scores [11].

Wu et al. put forward a subjective acoustic feature enhancement that relies on GAN-based architectures. Their system has significantly improved correlation of arousal and dominance, and thus it is very well suited for noisy speech enhancement [12].

III. METHODOLOGY

A. System Overview

The proposed model processes raw audio input and produces two outputs: explicit emotion prediction and hidden emotion score. Speech signals are first preprocessed and passed through the HuBERT-based encoder. The encoder output is then refined through self-supervised adapter layers before being fed into a dual-path attention module that generates separate explicit and hidden emotion predictions. The KL divergence between these two predictions is the suppression score.

B. Data Collection and Preprocessing

We used two publicly accessible datasets for emotional speech recognition, taken from the Kaggle platform.

IEMOCAP - Intensive Emotional Daily Motion Capture
The IEMOCAP dataset has become one of the most popular

datasets in terms of automatic analysis of speech emotions and consists of about 12 hours of dyadic conversation collected from ten actors in five sessions. Their speech expressions have been annotated with human labeling.

CREMA-D - Crowd-Sourced Emotional Multimodal Actors Dataset

CREMA-D is an extensive multimodal dataset consisting of 7,442 speech segments collected from 10 actors of different

(i) Stage 1: Self-Supervised Adapter Training

The encoder weights from Stage 1 are frozen, and only the adapter parameters are trained. A lightweight bottleneck adapter layer is introduced into each transformer block of the HuBERT encoder. This parameter-efficient design reduces computational cost while preserving the rich speech representations acquired during pre-training. The adapters are trained on the CREMA-D dataset using a masked token

if encoder layers is above the selected context. This is the encoder's



0% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Submitted works
- Internet sources
- Crossref database
- Crossref posted content database

Match Groups

-  **0 Not Cited or Quoted 0%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 0%  Internet sources
- 0%  Publications
- 0%  Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- 
0 Not Cited or Quoted 0%
 Matches with neither in-text citation nor quotation marks
 - 
0 Missing Quotations 0%
 Matches that are still very similar to source material
 - 
0 Missing Citation 0%
 Matches that have quotation marks, but no in-text citation
 - 
0 Cited and Quoted 0%
 Matches with in-text citation present, but no quotation marks
-

Top Sources

- 0%  Internet sources
- 0%  Publications
- 0%  Submitted works (Student Papers)